

COMP4141 Slides Week 10

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Admin

Sample Exam

- Has been released
- Solutions will be provided later
- Feel free to discuss with me :)

Feedback From Marking

- Make sure to mention the reduction is polynomial time when trying to prove NP-hardness.
- Be clear on what your verifier is when you try to use one.
- Make sure to explain why your algorithm/verifier is polynomial time in NP inclusion proofs.

W9 P5

Problem 5

Consider the following decision problem

2-SAT

Input: A boolean formula φ in 2-CNF (at most 2 literals per clause)

Question: Is φ satisfiable?

- Last week I made a mistake; the condition for non-satisfiability is that there is a path from $\neg y$ to y and from y to $\neg y$ (i.e. bidirectional reachability).
- The same idea, using that NL is closed under complement, shows that we can find such paths for an unsatisfiable formula.
- If you know more graph theory, this bidirectional reachability means that y and $\neg y$ lie in the same strongly connected component (SCC).

PP

Probabilistic Algorithms

Probabilistic Algorithms

A probabilistic TM (PTM) M is a type of NTM in which each non-deterministic step is called a *coin-flip step* and has two legal next moves.

The probability of a branch b of M 's computation on input w is

$$\Pr[b] = \frac{1}{2^k},$$

where k is the number of coin-flip steps that occur on b .

The acceptance probability is the sum of the probabilities of all accepting branches (the rejection probability is its complement).

PP

PP

Say that a PTM M probabilistically recognises A if it runs in polynomial time and:

1. $w \in A \implies \Pr[M \text{ accepts } w] > \frac{1}{2}$,
2. $w \notin A \implies \Pr[M \text{ rejects } w] > \frac{1}{2}$.

Define **PP** as the class of languages that are probabilistically recognised by polynomial-time PTMs.

Working with **PP**

For example, $\text{SAT} \in \mathbf{PP}$.

PP in the hierarchy

$\mathbf{NP} \subseteq \mathbf{PP} \subseteq \text{PSPACE}$

Bounded-error Probabilistic Polynomial Time

BPP

BPP

For $\varepsilon \in [0, \frac{1}{2})$, a PTM M recognises A with error probability ε if:

1. $w \in A \implies \Pr[M \text{ accepts } w] \geq 1 - \varepsilon$,
2. $w \notin A \implies \Pr[M \text{ rejects } w] \geq 1 - \varepsilon$.

BPP is the class of languages recognised by polynomial-time PTMs with error probability $\frac{1}{3}$.

BPP example

$\text{PRIMES} = \{n \mid n \text{ is prime (in binary)}\}$

$\text{PRIMES} \in \mathbf{BPP}$

Amplification Lemma

Amplification Lemma

Let $\varepsilon \in [0, \frac{1}{2})$ and let $p(n) > 0$ be a polynomial. If a polynomial-time PTM M recognises A with error probability ε , then there exists another polynomial-time PTM M' that also recognises A with error probability $\exp(-p(n))$.

Interpretation

The definition of **BPP** is robust with respect to the choice of error probability; $\frac{1}{3}$ is just one convenient choice in $[0, \frac{1}{2})$.

RP & ZPP

RP

RP

RP is the class of languages recognisable by probabilistic polynomial-time TMs such that:

- If $w \in A$, then $\Pr[M \text{ accepts } w] \geq \frac{1}{2}$,
- If $w \notin A$, then $\Pr[M \text{ rejects } w] = 1$.

(Also called Monte Carlo PTIME.)

$$\mathbf{coRP} = \{A \mid \overline{A} \in \mathbf{RP}\}, \quad \mathbf{ZPP} = \mathbf{RP} \cap \mathbf{coRP}.$$

Results in **RP**

Lemma

$\text{RP} \subseteq \text{BPP}$

$\text{RP} \subseteq \text{NP}$

Las Vegas Algorithms

Las Vegas

A probabilistic TM M is called *Las Vegas* if for all $w \in \Sigma^*$:

$$\Pr(M \text{ accepts } w) \in \{0, 1\},$$

and for some polynomial p , the expected running time of M on inputs of length n is at most $p(n)$.

Las Vegas and ZPP

ZPP is the class of languages recognised by Las Vegas algorithms.

ZPP Problem

ZPP characterisation

Show that **ZPP** can be characterised as the set of languages L such that there exists a probabilistic Turing machine which:

- Runs in polynomial time and halts in ACCEPT, REJECT, or DON'T KNOW,
- On input $x \in L$, halts in ACCEPT with probability $p > \frac{1}{2}$ or in DON'T KNOW,
- On input $x \notin L$, halts in REJECT with probability $p' > \frac{1}{2}$ or in DON'T KNOW.

ZPP Solution

If $L \in \mathbf{ZPP} = \mathbf{RP} \cap \mathbf{coRP}$, do the following on input w :

- Run w on the **RP** machine. If it accepts, then ACCEPT,
- Otherwise run w on the **coRP** machine. If it rejects, then REJECT,
- Otherwise output DON'T KNOW.

Conversely, suppose we have such a machine:

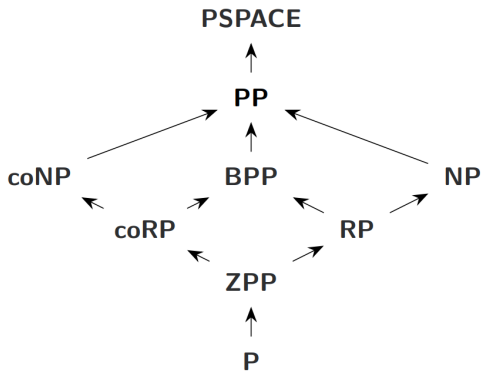
- If we output REJECT on DON'T KNOW, we obtain an **RP** algorithm,
- If we output ACCEPT on DON'T KNOW, we obtain a **coRP** algorithm.

Alternatively, to obtain a Las Vegas algorithm:

- On output DON'T KNOW, rerun the machine.

This yields finite expected running time.

Relationships



W10 P1

Problem 1

Probabilistic computers use a fair coin-flip to make probabilistic choices, is this enough to cover any probability distribution?

(a) Show how a fair coin can simulate rolling a dice.

Problem 1

Probabilistic computers use a fair coin-flip to make probabilistic choices, is this enough to cover any probability distribution?

- (b) Show how a fair coin can simulate making a choice with probability p for any $p \in (0, 1)$.

W10 P2

Problem 2

In the lecture, we showed that $\text{SAT} \in \mathbf{PP}$ and concluded that $\mathbf{NP} \subseteq \mathbf{PP}$. There is actually a missing step in the argument, relating to how $\mathbf{PP}$ interacts with polynomial time reductions. Formulate and prove a result about this, and use it to complete the argument.

W10 P3

Problem 3

Prove or disprove: **BPP** is closed under intersection. That is, if $A \in \mathbf{BPP}$ and $B \in \mathbf{BPP}$ then $A \cap B \in \mathbf{BPP}$.

W8 P5

Problem 5

Consider the following problem:

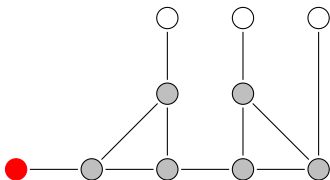
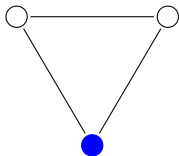
3-CoL

Input: An undirected graph G

Question: Is there a colouring of the vertices of G using at most 3 colours such that no edge has both endpoints with the same colour?

Problem 5

- (a) What can be said about the colouring of the white vertices in the following (partially coloured) graphs?



Problem 5

(b) Show that 3-CoL is **NP**-complete, assuming that SAT is **NP**-complete.

W8 P6

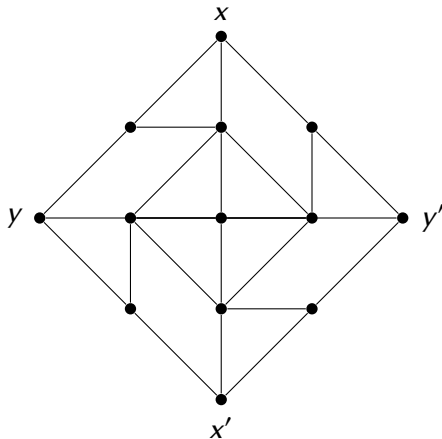
Problem 6

Show that if $\mathbf{P} = \mathbf{NP}$, then every language $A \in \mathbf{P}$ except $A = \emptyset$ and $A = \Sigma^*$, is $\mathbf{NP}$ -complete.

W8 Bonus

Bonus Week 8

Consider the following graph:



Bonus Week 8

(a) Give a valid 3-colouring of this graph where x and y have different colours.

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- (a) Give a valid 3-colouring of this graph where x and y have different colours.
- (b) Give a valid 3-colouring of this graph where x and y have the same colour.

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- (b) Give a valid 3-colouring of this graph where x and y have the same colour.
- (c) What can be said about *any* valid 3-colouring of this graph? Justify your answer.

Bonus Week 8

- (a) Give a valid 3-colouring of this graph where x and y have different colours.
- (b) Give a valid 3-colouring of this graph where x and y have the same colour.
- (c) What can be said about *any* valid 3-colouring of this graph? Justify your answer.
- (d) Consider the following decision problem:

PLANAR 3-COL

Input: A planar graph G

Question: Does G have a valid 3-colouring?

Show that PLANAR 3-COL is NP-complete.

Hint: Use the above graph to “uncross” edges.

Bonus Week 8

- (e) Prove that there is no planar graph G , with x, y, x', y' on an outer face (in that order), that satisfies the conditions indicated in (a), (b) and (c), but with a 4-colouring rather than a 3-colouring.